✪ Psychological Statistics ✪

Mixed ANOVA (repeated-measure & N-way)

- Edited by Prof. **Changwei Wu**
- Graduate Institute of Mind, Brain and Consciousness (***GIMBC***), Taipei Medical University

In [ ]:
```
### [ Setup the working directory ]

setwd("/Users/wesley/[Course]/Python/R_Script")
getwd()
```

In [23]:
```
### [ Loading the required libraries ]

library("dplyr")
library("rstatix")
library("ggpubr")
library("pastecs")
library("multcomp")
library("compute.es")
library("broom")
```

---

(1) One-way Repeated-measure ANOVA

Example: Bushtucker

[Hypothesis] Rat growth can be predicted by dietary tannin concentration. *(2-tailed)*

- Null hypothesis H_0 **:** No linear relationship between rat growth and tannin concentration $\rightarrow \beta = 0$
- Alternative hyp. H_1 **:** There is (linear) relationship between rat growth and tannin concentration $\rightarrow \beta \neq 0$

In [19]:
```
### [ Step.1 ] Load data

bushData <- read.delim("Bushtucker.dat", header = TRUE)

longBush <- bushData %>% gather(key = "Animal",
            value = "Retch", "stick_insect","kangaroo_testicle","fish_eye","wit
# ---(same function)---
# longBush <- reshape2::melt(bushData, id = "participant",
#             measured = c("stick_insect","kangaroo_testicle","fish_eye","witch

# ---(rename the labels in Animal)
names(longBush) <- c("Participant", "Animal", "Retch")
longBush$Animal <- factor(longBush$Animal,
                labels = c("Stick Insect","Kangaroo Ball","Fish Eye","Witchetty
#longBush <- longBush[order(longBush$Participant),]

head(longBush)
```

A data.frame: 6 × 3

	Participant	Animal	Retch
	<fct>	<fct>	<int>
1	P1	Fish Eye	8
2	P2	Fish Eye	9
3	P3	Fish Eye	6
4	P4	Fish Eye	5
5	P5	Fish Eye	8
6	P6	Fish Eye	7

In [16]:
```
### [ Step.2 ] Check assumptions

#----- (a) Outliers -----#
longBush %>%
    group_by(Animal) %>%
    identify_outliers(Retch)

#----- (b) Normality -----#
longBush %>%
    group_by(Animal) %>%
    shapiro_test(Retch)

#----- (c) Homogeneity of Variance -----#
longBush %>%
    levene_test(Retch ~ Animal)
```

A data.frame: 0 × 5

Animal	participant	Retch	is.outlier	is.extreme
<fct>	<fct>	<int>	<lgl>	<lgl>

A tibble: 4 × 4

Animal	variable	statistic	p
<fct>	<chr>	<dbl>	<dbl>
Stick Insect	Retch	0.9126421	0.3730451
Kangaroo Ball	Retch	0.9388881	0.6001739
Fish Eye	Retch	0.9815777	0.9702866
Witchetty Grub	Retch	0.9005734	0.2924091

A tibble: 1 × 4

df1	df2	statistic	p
<int>	<int>	<dbl>	<dbl>
3	28	0.8422764	0.4822894

In [50]:
```
### [ Step.3-1 ] repeated-measure ANOVA

(bushModel <- anova_test(data = longBush, dv = Retch, wid = Participant,
                         within = Animal, detailed=TRUE))
```

$ANOVA

A data.frame: 2 × 9

Effect	DFn	DFd	SSn	SSd	F	p	p<.05	ges
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
(Intercept)	1	7	990.1	17.4	398.90	1.97e-07	*	0.853
Animal	3	21	83.1	153.4	3.79	2.60e-02	*	0.327

$`Mauchly's Test for Sphericity`

A data.frame: 1 × 4

	Effect	W	p	p<.05
	<chr>	<dbl>	<dbl>	<chr>
1	Animal	0.136	0.047	*

$`Sphericity Corrections`

A data.frame: 1 × 9

Effect	GGe	DF[GG]	p[GG]	p[GG]<.05	HFe	DF[HF]	p[HF]	p[HF]<.05
<chr>	<dbl>	<chr>	<dbl>	<chr>	<dbl>	<chr>	<dbl>	<chr>
Animal	0.533	1.6, 11.19	0.063		0.666	2, 13.98	0.048	*

In [25]:
```
bushModelaov <- aov(Retch ~ Animal + Error(Participant/Animal), data = longBush
summary(bushModelaov)
```

```
Error: Participant
          Df Sum Sq Mean Sq F value Pr(>F)
Residuals  7  17.38   2.482

Error: Participant:Animal
          Df Sum Sq Mean Sq F value Pr(>F)
Animal     3  83.13  27.708   3.794 0.0256 *
Residuals 21 153.38   7.304
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

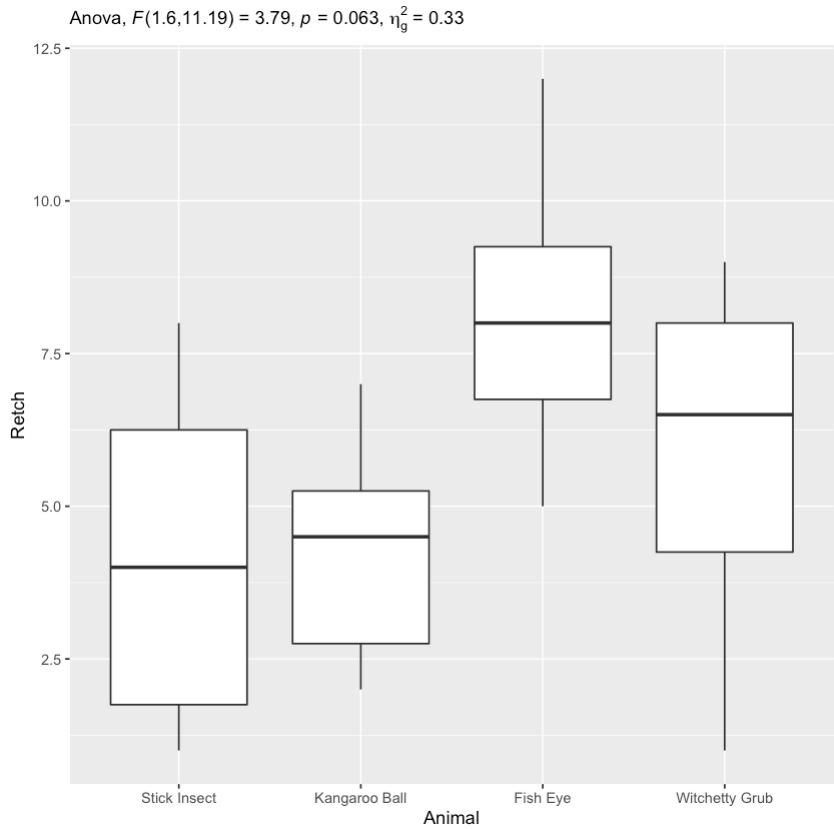
In [26]:
```
### [ Step.3-2 ] Post-hoc Tests

#--- Pairwise comparisons ---#
pairwise.t.test(longBush$Retch, longBush$Animal,
                paired = TRUE, p.adjust.method = "bonferroni")
```

```
        Pairwise comparisons using paired t tests

data:  longBush$Retch and longBush$Animal

               Stick Insect Kangaroo Ball Fish Eye
Kangaroo Ball  1.0000       -             -
Fish Eye       0.0056       0.0121        -
Witchetty Grub 1.0000       1.0000        1.0000

P value adjustment method: bonferroni
```

In [27]:
```
### [ Step.4 ] Effect size

chi2fromMauchlysTest <-function(model)
{
    a <- data.frame(model[1])[,2]
    b <- data.frame(model[1])[,3]
    c <- data.frame(model[2])[,2]
    p <- data.frame(model[2])[,3]
    dof <- a[length(a)]

    # number of repeated measures
    k <- dof+1
    # number of participants
    n <- b[1]+1
    d <- 1 - ((2 * ((k - 1)^2)+(k - 1)+2)/(6*(k - 1)*(n - 1)))
    # W statitic
    W <- c[length(c)] #Mauchly's W from anova_test
    pval <- p[length(p)]
    chi <- -(n-1)*d*log(W)      # approximate chi^2
    df <- (k*(k - 1)/2) - 1    # df for chi^2
    print(paste("Mauchly's Test for Sphericity:"))
    print(paste("W statistic = : ", W, " and corresponding p-value = ", pval),
    print(paste("Approximate Chi-Square = ", chi, " with DOF =", df), digits=3)
}

chi2fromMauchlysTest(bushModel)
```

```
[1] "Mauchly's Test for Sphericity:"
[1] "W statistic = :  0.136  and corresponding p-value =  0.047"
[1] "Approximate Chi-Square =  11.416407805797  with DOF = 5"
```

In [28]:
```
### [ Step.5 ] Visualization

ggplot(longBush, aes(Animal, Retch)) +
    geom_boxplot()+
    labs(subtitle = get_test_label(bushModel, detailed = TRUE))
```



~ *Report* ~

- A repeated-measure ANOVA shows that the time to retch was not significantly affected by the type of eaten animal, $\boldsymbol{F_{1.60,\ 11.19} = 3.79,\ p > .05}$, generalized η^2 = . 33.
- Mauchly's test indicated that the assumption of sphericity had been violated, $\chi^2_{(5)} = 11.41$, $p < .05$, therefore Greenhouse-Geisser corrected tests are reported (ε = .53).

---

(2) Two-way Independent ANOVA

Example A: Beer-Goggle Effect

[Hypothesis] Subjective perception of physical attractiveness would become more inaccurate after alcohol consumption. *(2-tailed)*

- Null hypothesis $\boldsymbol{H_0}$ **:** No prominent interactions can be observed between Gender and Alcohol effects.
- Alternative hyp. $\boldsymbol{H_1}$ **:** There is significant interactions between Gender and Alcohol effects.

In [31]:
```
### [ Step.1 ] Data Loading

# → Loading the dataset
gogglesData<-read.csv("goggles.csv", header = TRUE)
gogglesData$alcohol<-factor(gogglesData$alcohol, levels = c("None", "2 Pints",

gogglesData %>% head(5)
```

A data.frame: 5 × 3

	gender	alcohol	attractiveness
	<fct>	<fct>	<int>
1	Female	None	65
2	Female	None	70
3	Female	None	60
4	Female	None	60
5	Female	None	60

In [33]:
```
### [ Step.2 ] Check assumptions

#----- (a) Outliers -----#
gogglesData %>%
    group_by(gender, alcohol) %>%
    identify_outliers(attractiveness)

#----- (b) Normality -----#
gogglesData %>%
    group_by(gender, alcohol) %>%
    shapiro_test(attractiveness)

#----- (c) Homogeneity of Variance -----#
gogglesData %>%
    group_by(gender) %>%
    levene_test(attractiveness ~ alcohol)
```

A tibble: 2 × 5

gender	alcohol	attractiveness	is.outlier	is.extreme
<fct>	<fct>	<int>	<lgl>	<lgl>
Female	None	70	TRUE	TRUE
Female	2 Pints	50	TRUE	FALSE

A tibble: 6 × 5

gender	alcohol	variable	statistic	p
<fct>	<fct>	<chr>	<dbl>	<dbl>
Female	None	attractiveness	0.8715152	0.1559521
Female	2 Pints	attractiveness	0.8989639	0.2828089
Female	4 Pints	attractiveness	0.8972707	0.2729917
Male	None	attractiveness	0.9410603	0.6215419
Male	2 Pints	attractiveness	0.9666411	0.8704264
Male	4 Pints	attractiveness	0.9508657	0.7199577

A tibble: 2 × 5

gender	df1	df2	statistic	p
<fct>	<int>	<int>	<dbl>	<dbl>
Female	2	21	0.49606299	0.6158810
Male	2	21	0.08832808	0.9157988

In [51]:
```
### [ Step.3-1 ] ANOVA

gogglesModel<-anova_test(attractiveness ~ gender + alcohol + gender:alcohol,
                         data = gogglesData, detailed = TRUE)
gogglesModel
```

```
Coefficient covariances computed by hccm()
```

A anova_test: 3 × 9

	Effect	SSn	SSd	DFn	DFd	F	p	p<.05	ges
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	gender	169	3488	1	42	2.03	1.61e-01		0.046
2	alcohol	3332	3488	2	42	20.07	7.65e-07	*	0.489
3	gender:alcohol	1978	3488	2	42	11.91	7.99e-05	*	0.362

In [52]:
```
### [ Step.3-2 ] post-hoc test

#----- Significant interaction → 1. Break into simple main effects -----#
gogglesData %>%
    group_by(gender) %>%
    anova_test(attractiveness ~ alcohol, detailed = TRUE)

#-------- 2. simple pairwise comparison --------
pwc <- gogglesData %>%
    filter(gender=='Male') %>%
    pairwise_t_test(attractiveness~alcohol, p.adjust.method = "bonferroni")
pwc
```

```
Coefficient covariances computed by hccm()

Coefficient covariances computed by hccm()
```

A grouped_anova_test: 2 × 10

	gender	Effect	SSn	SSd	DFn	DFd	F	p	p<.05	ges
	<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	Female	alcohol	102	822	2	21	1.3	2.92e-01		0.110
2	Male	alcohol	5208	2666	2	21	20.5	1.15e-05	*	0.661

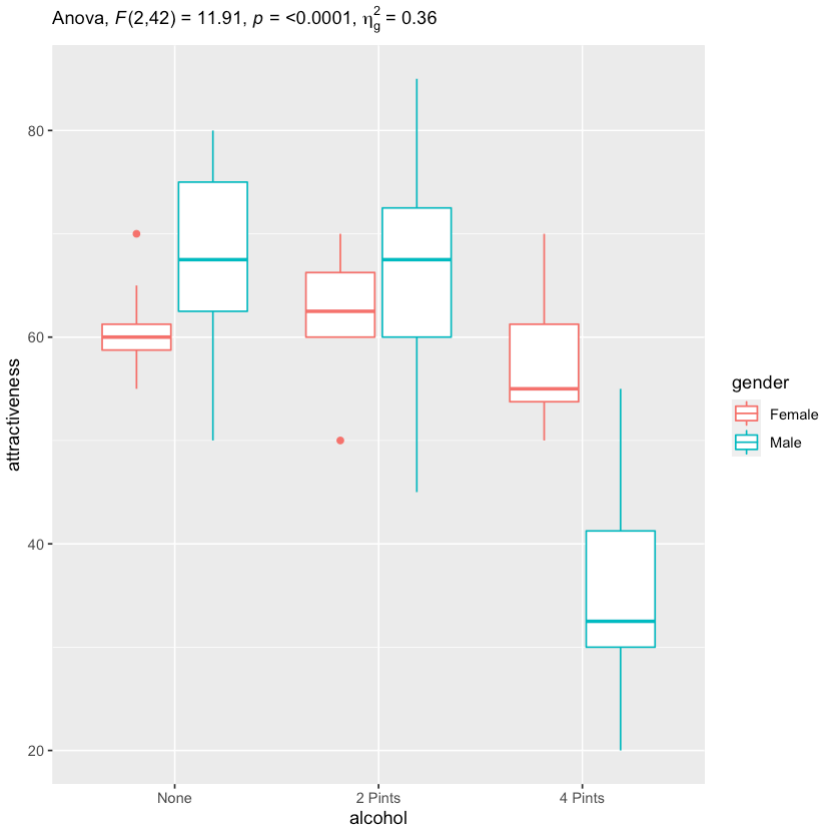
A rstatix_test: 3 × 9

	.y.	group1	group2	n1	n2	p	p.signif	p.adj	p.adj.signif
	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<chr>	<dbl>	<chr>
1	attractiveness	None	2 Pints	8	8	1.00e+00	ns	1e+00	ns
2	attractiveness	None	4 Pints	8	8	1.67e-05	****	5e-05	****
3	attractiveness	2 Pints	4 Pints	8	8	1.67e-05	****	5e-05	****

In [38]:
```
### [ Step.4 ] Effect size

#-------- Effect size --------
omega_factorial<-function(n,a,b, SSa, SSb, SSab, SSr)
{
    MSa<-SSa/(a-1)
    MSb<-SSb/(b-1)
    MSab<-SSab/((a-1)*(b-1))
    MSr<-SSr/(a*b*(n-1))
    varA<-((a-1)*(MSa-MSr))/(n*a*b)
    varB<-((b-1)*(MSb-MSr))/(n*a*b)
    varAB<-((a-1)*(b-1)*(MSab-MSr))/(n*a*b)
    varTotal<-varA + varB + varAB + MSr

    print(paste("Omega-Squared A: ", varA/varTotal),digits=3)
    print(paste("Omega-Squared B: ", varB/varTotal),digits=3)
    print(paste("Omega-Squared AB: ", varAB/varTotal),digits=3)
}

omega_factorial(48,2,3,169,3332,1978,3488)


#Gender effects (within alcohol)
#library(compute.es)
#mes(66.875, 60.625, 10.3293963, 4.95515604, 8, 8)
#mes(66.875, 62.5, 12.5178444, 6.5465367, 8, 8)
#mes(35.625, 57.5, 10.8356225, 7.0710678, 8, 8)
```

```
[1] "Omega-Squared A:  0.0174434539065517"
[1] "Omega-Squared B:  0.368317917116542"
[1] "Omega-Squared AB:  0.217527808032756"
```

In [39]:
```
### [ Step.5 ] Visualization

ggplot(gogglesData, aes(alcohol, attractiveness)) +
    geom_boxplot(aes(color=gender))+
    labs(subtitle = get_test_label(gogglesModel, detailed = TRUE))
```



~ *Report* ~

- **[Interaction]** There was a significant interaction effect between the amount of alcohol consumed and the gender of the person selecting a mate, on the attractiveness of the partner selected, $F_{2, 42} = 11.9, p < .001, \omega^2 = .22$. This indicates that male and female genders were affected differently by alcohol.
- **[Post-hoc]** There was a significant simple main effect of the amount of alcohol consumption lowering the attractiveness of the male group only, $F_{2, 21} = 20.5, p < .001$. Pairwise comparison Bonferroni revealed that attractiveness was lower after 4 pints than both after 2 pints and no alcohol in the male group ($p < .001$). The attractiveness of males after 2 pints and no alcohol were not significantly different.

Example B - 2-way Repeated-measure ANOVA

[Hypothesis] Comparing the effects of negative photos against positive/neutral photos after consuming different types of drinks. *(2-tailed)*

- Null hypothesis H_0 **:** No prominent interactions can be observed between drink types and emotion types.
- Alternative hyp. H_1 **:** Significant interactions can be observed between drink types and emotion types.

In [41]:
```
### [ Step.1 ] Data Loading

# → Loading the dataset
attitudeData<-read.delim("Attitude.dat", header = TRUE)
longAttitude <-reshape2::melt(attitudeData, id = "participant",
                    measured = c( "beerpos", "beerneg", "beerneut",
                                  "winepos", "wineneg", "wineneut",
                                  "waterpos", "waterneg", "waterneu"))

names(longAttitude)<-c("participant", "groups", "attitude")

longAttitude$drink<-gl(3, 60, labels = c("Beer", "Wine", "Water"))
longAttitude$emotion<-gl(3,20, 180, labels = c("Positive", "Negative", "Neutral
longAttitude<-longAttitude[order(longAttitude$participant),]

longAttitude %>% head(5)
```

A data.frame: 5 × 5

	participant	groups	attitude	drink	emotion
	<fct>	<fct>	<int>	<fct>	<fct>
1	P1	beerpos	1	Beer	Positive
21	P1	beerneg	6	Beer	Negative
41	P1	beerneut	5	Beer	Neutral
61	P1	winepos	38	Wine	Positive
81	P1	wineneg	-5	Wine	Negative

In [43]:
```
### [ Step.2 ] Check assumptions

options(digits = 3)

#----- (a) Outliers -----#
longAttitude %>%
    group_by(drink, emotion) %>%
    identify_outliers(attitude)

#----- (b) Normality -----#
longAttitude %>%
    group_by(drink, emotion) %>%
    shapiro_test(attitude)

#----- (c) Homogeneity of Variance -----#
#[longAttitude %>% group_by(drink) %>% levene_test(attitude ~ emotion)]
# No need to check the assumption of Variance Homogeneity,
# because all factors are within-subject (repeated measure).
```

A tibble: 3 × 7

drink	emotion	participant	groups	attitude	is.outlier	is.extreme
<fct>	<fct>	<fct>	<fct>	<int>	<lgl>	<lgl>
Wine	Positive	P5	winepos	11	TRUE	FALSE
Water	Neutral	P2	waterneu	-13	TRUE	FALSE
Water	Neutral	P6	waterneu	-13	TRUE	FALSE

A tibble: 9 × 5

drink	emotion	variable	statistic	p
<fct>	<fct>	<chr>	<dbl>	<dbl>
Beer	Positive	attitude	0.951	0.3751
Beer	Negative	attitude	0.896	0.0354
Beer	Neutral	attitude	0.961	0.5728
Wine	Positive	attitude	0.987	0.9907
Wine	Negative	attitude	0.954	0.4300
Wine	Neutral	attitude	0.941	0.2523
Water	Positive	attitude	0.970	0.7537
Water	Negative	attitude	0.970	0.7614
Water	Neutral	attitude	0.909	0.0620

In [53]:
```
### [ Step.3-1 ] ANOVA

(attitudeModel<-anova_test(data = longAttitude, dv = attitude,
                           wid = participant,  within = c(drink, emotion), deta

attitudeModel.aov<-aov(attitude ~ (drink * emotion) +
                       Error(participant/(drink * emotion)), data = longAttitu
summary(attitudeModel.aov)
```

$ANOVA

A data.frame: 4 × 9

Effect	DFn	DFd	SSn	SSd	F	p	p<.05	ges
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
(Intercept)	1	19	11218	1920	111.00	2.26e-09	*	0.413
drink	2	38	2092	7786	5.11	1.10e-02	*	0.116
emotion	2	38	21629	3353	122.56	2.68e-17	*	0.575
drink:emotion	4	76	2624	2907	17.16	4.59e-10	*	0.141

$`Mauchly's Test for Sphericity`

A data.frame: 3 × 4

	Effect	W	p	p<.05
	<chr>	<dbl>	<dbl>	<chr>
1	drink	0.267	6.95e-06	*
2	emotion	0.662	2.40e-02	*
3	drink:emotion	0.595	4.36e-01	

$`Sphericity Corrections`

A data.frame: 3 × 9

Effect	GGe	DF[GG]	p[GG]	p[GG]<.05	HFe	DF[HF]	p[HF]	p[HF]<.05
<chr>	<dbl>	<chr>	<dbl>	<chr>	<dbl>	<chr>	<dbl>	<chr>
drink	0.577	1.15, 21.93	3.00e-02	*	0.591	1.18, 22.45	2.90e-02	*
emotion	0.747	1.49, 28.4	1.76e-13	*	0.797	1.59, 30.28	3.14e-14	*
drink:emotion	0.798	3.19, 60.68	1.90e-08	*	0.979	3.91, 74.37	6.81e-10	*

```
Error: participant
          Df Sum Sq Mean Sq F value Pr(>F)
Residuals 19   1920     101

Error: participant:drink
          Df Sum Sq Mean Sq F value Pr(>F)
drink      2   2092    1046    5.11  0.011 *
Residuals 38   7786     205
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Error: participant:emotion
          Df Sum Sq Mean Sq F value Pr(>F)
emotion    2  21629   10814     123 <2e-16 ***
Residuals 38   3353      88
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Error: participant:drink:emotion
              Df Sum Sq Mean Sq F value  Pr(>F)
drink:emotion  4   2624     656    17.1 4.6e-10 ***
Residuals     76   2907      38
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

In [57]:
```
### [ Step.3-2 ] Post-hoc tests

#-> 1. Simple main effect: drink effects at each emotion
drink.effect <- longAttitude %>%
    group_by(emotion) %>%
    anova_test(dv = attitude, wid = participant,  within = drink)

get_anova_table(drink.effect) %>%
    adjust_pvalue(method='bonferroni')

# longAttitude %>% filter(emotion == 'Negative') %>%
#    anova_test(dv = attitude, wid = participant,  within = drink, type = 2)
# longAttitude %>% filter(emotion == 'Neutral') %>%
#    anova_test(dv = attitude, wid = participant,  within = drink, type = 2)

#-> 2. Pairwise comparison
(pwc <- longAttitude %>%
    group_by(emotion) %>%
    filter(emotion %in% c('Negative', 'Neutral')) %>%
    pairwise_t_test(attitude ~ drink, paired = TRUE, p.adjust.method="bonferron
```

A tibble: 3 × 9

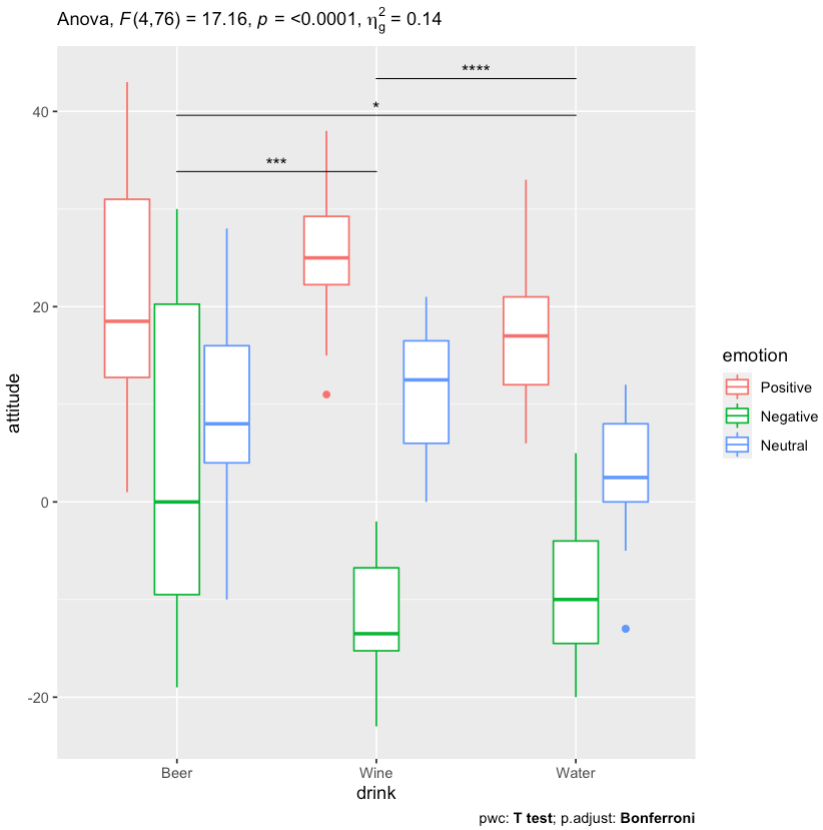
emotion	Effect	DFn	DFd	F	p	p<.05	ges	p.adj
<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>
Positive	drink	1.55	29.5	3.06	0.074000		0.112	0.22200
Negative	drink	1.22	23.3	14.08	0.000555	*	0.298	0.00167
Neutral	drink	1.33	25.2	7.27	0.008000	*	0.213	0.02400

A rstatix_test: 6 × 11

	emotion	.y.	group1	group2	n1	n2	statistic	df	p	p.adj	p.adj.signif
	<fct>	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	Negative	attitude	Beer	Wine	20	20	5.035	19	7.36e-05	2.21e-04	***
2	Negative	attitude	Beer	Water	20	20	3.153	19	5.00e-03	1.60e-02	*
3	Negative	attitude	Wine	Water	20	20	-1.479	19	1.55e-01	4.65e-01	ns
4	Neutral	attitude	Beer	Wine	20	20	-0.545	19	5.92e-01	1.00e+00	ns
5	Neutral	attitude	Beer	Water	20	20	2.521	19	2.10e-02	6.20e-02	ns
6	Neutral	attitude	Wine	Water	20	20	6.663	19	2.26e-06	6.78e-06	****

In [ ]:
```
### [ Step.4 ] Effect size

# Using the generalized η2 in factorial ANOVA → 0.141
```

In [58]:
```
### [ Step.5 ] Visualization

pwc <- pwc %>% add_xy_position(x="drink")
ggplot(longAttitude, aes(drink, attitude)) +
    geom_boxplot(aes(color=emotion)) +
    stat_pvalue_manual(pwc, tip.length = 0, hide.ns=TRUE) +
    labs(subtitle = get_test_label(attitudeModel, detailed = TRUE),
         caption = get_pwc_label(pwc))
```



~ *Report* ~

- **[Interaction]** There was a significant interaction effect between the type of drink and the type of imagery used, $\boldsymbol{F_{4,\,76} = 17.15,\ p < .001,\ \eta^2 = .141}$. This indicates that image emotion had different effects on people's ratings, depending on which type of drink was used.
- **[Post-hoc]** There was a significant main effect of drink on attitude ratings at different types of image emotions. The drink effect was significant at Negative photos ($\boldsymbol{F_{1.2,\,23.3} = 14.1,\ p < 0.001}$) and Neutral photos ($\boldsymbol{F_{1.3,\,25.2} = 7.27,\ p < 0.01}$) but not at Positive photos ($F_{1.6,\,29.5} = 3.06,\ p = 0.074$).
- **[Pairwise comparisons]** Pairwise tests revealed that for Negative emotion, there were significant differences between beer and both wine ($p < .001$) and water ($p = .016$), but not between wine and beer (p =.47); for Neutral, there were significant differences between water and wine ($p < .001$) only. These findings suggest that beer is unusual in that negative photo does appear to reduce attitudes compared to neutral imagery.

---

(3) Two-way ANCOVA

[Hypothesis] To evaluate the effect of treatment and exercise on stress reduction score after adjusting for age. *(2-tailed)*

In this example: 1) **stress score** is our outcome (dependent) variable; 2) **treatment** (levels: no and yes) and **exercise** (levels: low, moderate and high intensity training) are our grouping variable; 3) **age** is our covariate.

- Null hypothesis H_0 **:** After controling age, no interactions can be observed between treatment and exercise.
- Alternative hyp. H_1 **:** After controling age, there is significant interaction between treatment and exercise.

In [67]:
```
### [ Step.1 ] Data Loading

# → Loading the dataset
set.seed(123)
data("stress", package = "datarium")
stress %>% sample_n_by(treatment, exercise)

levels(stress$treatment)
levels(stress$exercise)

#----- Descriptive stat. -----
stress %>%
  group_by(treatment, exercise) %>%
  get_summary_stats(score, type = "mean_sd")
```

A tibble: 6 × 5

id	score	treatment	exercise	age
<int>	<dbl>	<fct>	<fct>	<dbl>
3	97.2	yes	low	70
13	78.9	yes	moderate	60
30	72.5	yes	high	58
32	96.1	no	low	62
46	91.1	no	moderate	60
55	75.3	no	high	52

'yes' · 'no'

'low' · 'moderate' · 'high'

A tibble: 6 × 6

treatment	exercise	variable	n	mean	sd
<fct>	<fct>	<chr>	<dbl>	<dbl>	<dbl>
yes	low	score	10	87.9	6.29
yes	moderate	score	10	86.8	5.69
yes	high	score	10	71.8	4.33
no	low	score	10	89.6	4.49
no	moderate	score	10	89.4	5.42
no	high	score	10	82.0	5.58

In [76]:
```
### [ Step.2 ] Check assumptions

#----- Linearity -----#
ggscatter(stress, x = "age", y = "score",
  facet.by  = c("exercise", "treatment"),
  short.panel.labs = FALSE)+
  stat_smooth(method = "loess", span = 0.9)

#----- (a) Normality -----#
# Build the linear model, the covariate goes first
model <- lm(score ~ age + treatment*exercise, data = stress)

# Inspect the model diagnostic metrics
model.metrics <- augment(model)
head(model.metrics, 3)

# Compute Shapiro-Wilk test of normality
shapiro_test(model.metrics$.resid)
# shapiro_test(residuals(model))

#----- (b) Outliers -----#
model.metrics %>%
  filter(abs(.std.resid) > 3) %>%
  as.data.frame()

#----- (c) Homogeneity of Variance -----#
levene_test(.resid ~ treatment*exercise, data = model.metrics)

#----- (d) Homogeneity of Regression Slope -----#
stress %>%
  anova_test(score ~ age + treatment + exercise + treatment*exercise
             + age*treatment + age*exercise + age*exercise*treatment)

# stress %>% unite(col = "group", treatment, exercise) %>% anova_test(score ~ g
```

```
`geom_smooth()` using formula 'y ~ x'
```

A tibble: 3 × 10

score	age	treatment	exercise	.fitted	.resid	.hat	.sigma	.cooksd	.std.resid
<dbl>	<dbl>	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
95.6	59	yes	low	86.5	9.10	0.108	4.85	0.0647	1.93
82.2	65	yes	low	89.5	-7.32	0.112	4.92	0.0439	-1.56
97.2	70	yes	low	92.0	5.16	0.177	4.97	0.0401	1.14

A tibble: 1 × 3

variable	statistic	p.value
<chr>	<dbl>	<dbl>
model.metrics$.resid	0.982	0.531

A data.frame: 0 × 10

score	age	treatment	exercise	.fitted	.resid	.hat	.sigma	.cooksd	.std.resid
<dbl>	<dbl>	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>

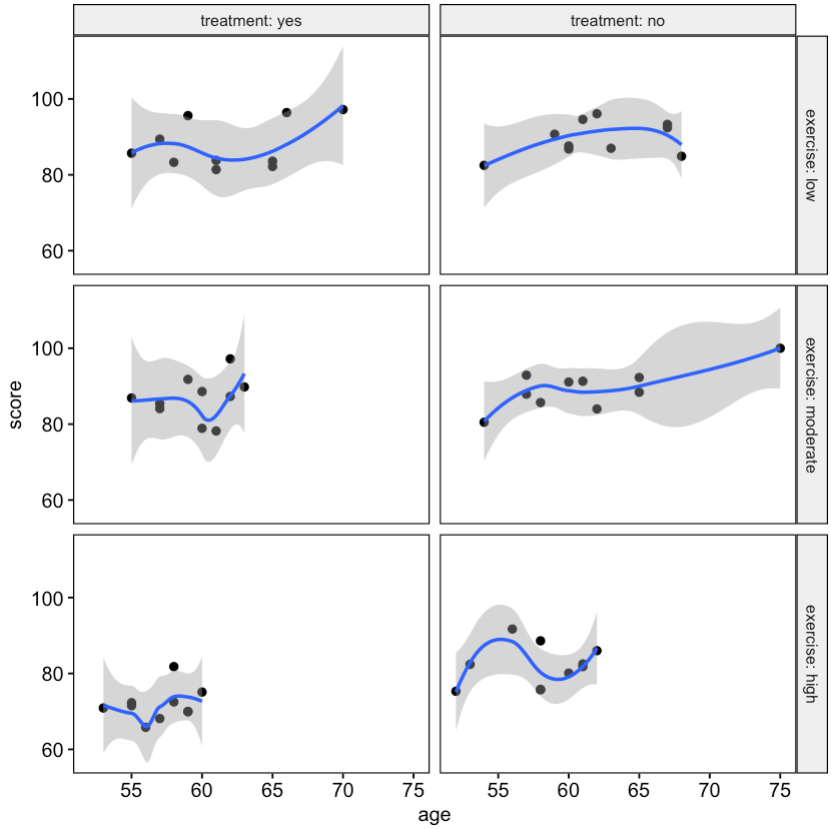
A tibble: 1 × 4

df1	df2	statistic	p
<int>	<int>	<dbl>	<dbl>
5	54	0.829	0.535

```
Coefficient covariances computed by hccm()
```

A anova_test: 7 × 7

	Effect	DFn	DFd	F	p	p<.05	ges
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	age	1	48	8.359	6.00e-03	*	0.148000
2	treatment	1	48	9.907	3.00e-03	*	0.171000
3	exercise	2	48	18.197	1.31e-06	*	0.431000
4	treatment:exercise	2	48	3.303	4.50e-02	*	0.121000
5	age:treatment	1	48	0.009	9.25e-01		0.000189
6	age:exercise	2	48	0.235	7.91e-01		0.010000
7	age:treatment:exercise	2	48	0.073	9.30e-01		0.003000



In [74]:
```
### [ Step.3-1 ] ANCOVA

res.aov <- stress %>% anova_test(score ~ age + treatment*exercise)
get_anova_table(res.aov)
```

```
Coefficient covariances computed by hccm()
```

A anova_test: 4 × 7

	Effect	DFn	DFd	F	p	p<.05	ges
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	age	1	53	9.11	4.00e-03	*	0.147
2	treatment	1	53	11.10	2.00e-03	*	0.173
3	exercise	2	53	20.82	2.13e-07	*	0.440
4	treatment:exercise	2	53	4.45	1.60e-02	*	0.144

In [77]:
```
### [ Step.3-2 ] Post-hoc test

#----- (a) Simple main effect: Effect of treatment at each level of exercise --
stress %>%
  group_by(exercise) %>%
  anova_test(score ~ age + treatment) %>%
  adjust_pvalue(method = 'bonferroni')

#----- (a) Simple pairwise comparisons -----
pwc <- stress %>%
  group_by(exercise) %>%
  emmeans_test(score ~ treatment, covariate = age,
               p.adjust.method = "bonferroni")
pwc %>% filter(exercise == "high")
```

```
Coefficient covariances computed by hccm()

Coefficient covariances computed by hccm()

Coefficient covariances computed by hccm()
```

A grouped_anova_test: 6 × 9

exercise	Effect	DFn	DFd	F	p	p<.05	ges	p.adj
<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>
low	age	1	17	2.252	0.152000		0.117	0.91200
low	treatment	1	17	0.437	0.517000		0.025	1.00000
moderate	age	1	17	6.647	0.020000	*	0.281	0.12000
moderate	treatment	1	17	0.419	0.526000		0.024	1.00000
high	age	1	17	0.794	0.385000		0.045	1.00000
high	treatment	1	17	18.742	0.000455	*	0.524	0.00273

A rstatix_test: 1 × 10

exercise	term	.y.	group1	group2	df	statistic	p	p.adj	p.adj.signif
<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
high	age*treatment	score	yes	no	53	-4.36	5.97e-05	5.97e-05	****

In [79]:
```
### [ Step.3-2 ] Post-hoc test

#----- (b) Simple main effect: Effect of exercise at each level of treatment --
stress %>%
  group_by(treatment) %>%
  anova_test(score ~ age + exercise)%>%
  adjust_pvalue(method = 'bonferroni')

#----- (b) Simple pairwise comparisons -----
pwc2 <- stress %>%
  group_by(treatment) %>%
  emmeans_test(score ~ exercise, covariate = age,
    p.adjust.method = "bonferroni")
pwc2 %>% filter(treatment == "yes")
```

```
Coefficient covariances computed by hccm()

Coefficient covariances computed by hccm()
```

A grouped_anova_test: 4 × 9

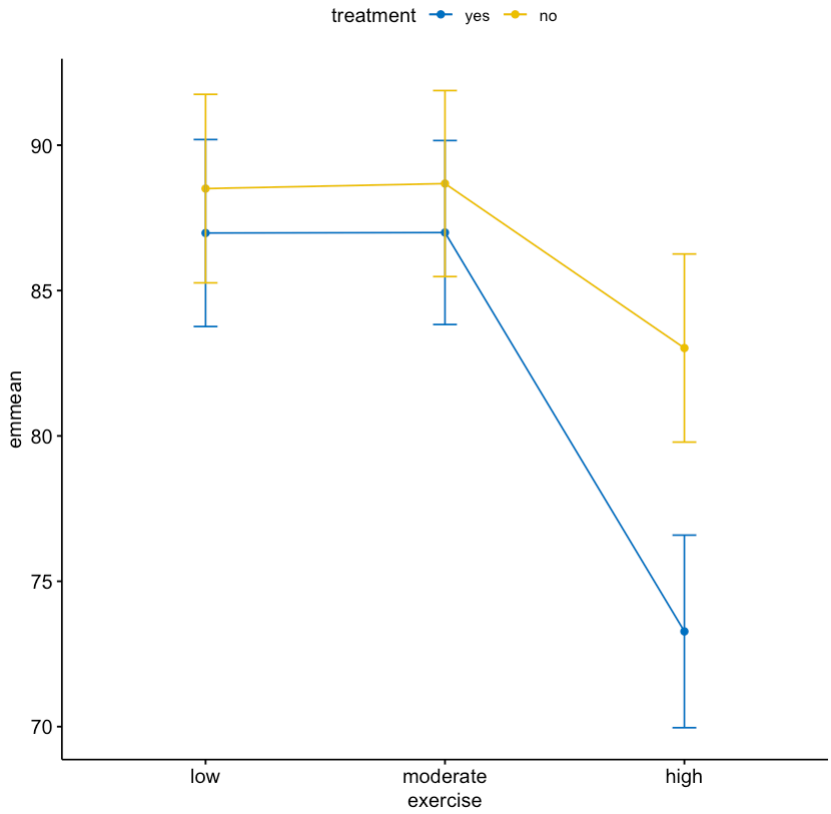
treatment	Effect	DFn	DFd	F	p	p<.05	ges	p.adj
<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>
yes	age	1	26	2.37	1.36e-01		0.083	5.44e-01
yes	exercise	2	26	17.34	1.64e-05	*	0.572	6.56e-05
no	age	1	26	7.26	1.20e-02	*	0.218	4.80e-02
no	exercise	2	26	3.99	3.10e-02	*	0.235	1.24e-01

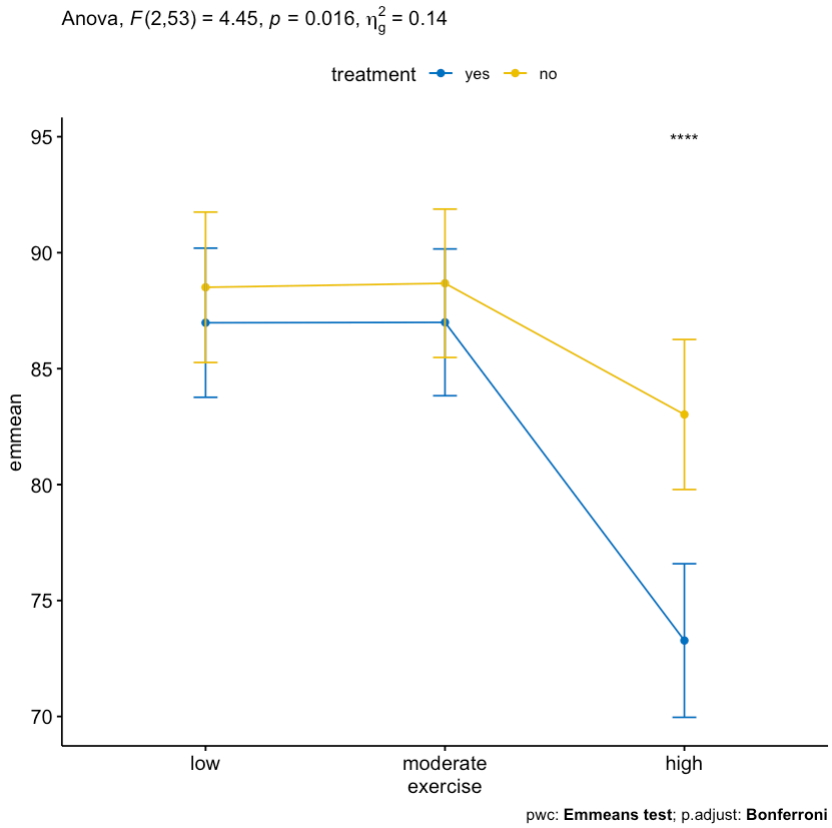
A rstatix_test: 3 × 10

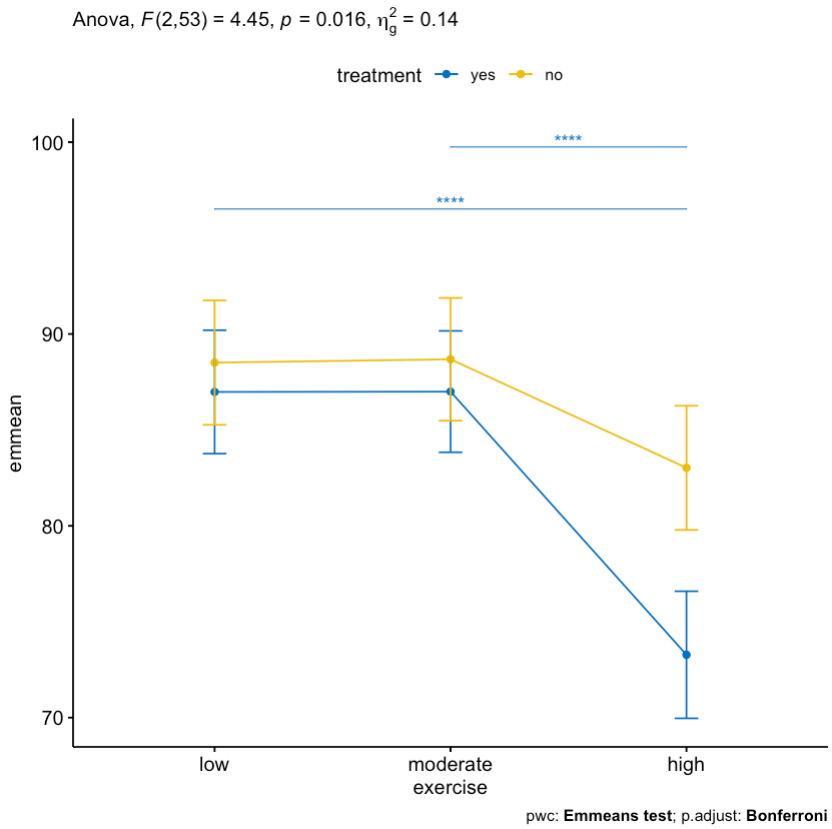
treatment	term	.y.	group1	group2	df	statistic	p	p.adj	p.adj.signif
<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
yes	age*exercise	score	low	moderate	53	-0.00774	9.94e-01	1.00e+00	ns
yes	age*exercise	score	low	high	53	5.79876	3.78e-07	1.13e-06	****
yes	age*exercise	score	moderate	high	53	6.04155	1.55e-07	4.66e-07	****

In [21]:
```
### [ Step.4 ] Effect size -- Generalized Eta Squared
```

In [80]:
```
### [ Step.5 ] Visualization

# Line plot
(lp <- ggline(get_emmeans(pwc), x = "exercise", y = "emmean", color = "treatmen
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high, color = treatment), widt

# Comparisons between treatment group at each exercise level
pwc <- pwc %>% add_xy_position(x = "exercise", fun = "mean_se", step.increase =
pwc.filtered <- pwc %>% filter(exercise == "high")
lp + stat_pvalue_manual(pwc.filtered, hide.ns = TRUE, tip.length = 0, bracket.s
  labs(subtitle = get_test_label(res.aov,  detailed = TRUE), caption = get_pwc_

# Comparisons between exercises group at each treatment level
pwc2 <- pwc2 %>% add_xy_position(x = "exercise", fun = "mean_se")
pwc2.filtered <- pwc2 %>% filter(treatment == "yes")
lp + stat_pvalue_manual(pwc2.filtered, hide.ns = TRUE, tip.length = 0,
    step.group.by = "treatment", color = "treatment") +
  labs(subtitle = get_test_label(res.aov,  detailed = TRUE),
    caption = get_pwc_label(pwc2))
```







~ *Report* ~

- A two-way ANCOVA was performed to examine the effects of treatment and exercise on stress reduction, after controlling for age.
- There was a significant two-way interaction between treatment and exercise on score concentration, whilst controlling for age, $\boldsymbol{F_{2,\,53} = 4.45,\ p = 0.016}$.
- Therefore, an analysis of simple main effects for exercise and treatment was performed with statistical significance receiving a Bonferroni adjustment and being accepted at the $p < 0.025$ level for exercise and $p < 0.017$ for treatment.
- The simple main effect of treatment was significant in the high-intensity exercise group ($p < 0.001$), but not in the low-intensity exercise group (p = 0.52) and the moderate-intensity exercise group (p =

0.53).

- The effect of exercise was significant in the treatment=yes group ($p < 0.001$), but not in the treatment=no group ($p = 0.031$).
- All pairwise comparisons were computed for significant simple main effects with reported p-values Bonferroni adjusted.
- For the treatment=yes group, there was a significant difference between the adjusted mean of low and high exercise group ($p < 0.001$) and, between moderate and high group ($p < 0.001$).
- The difference between the adjusted means of low and moderate exercise groups was not significant.

---

(4) Three-way Mixed ANOVA (1-Between / 2-Within)

[Hypothesis]* Looks or personality may be important for a speed-dating event, which may differ between gender.** ***(2-tailed)

- Null hypothesis H_0 **:** No prominent interactions can be observed between Looks, Personality and Gender.
- Alternative hyp. H_1 **:** Significant interactions can be observed between Looks, Personality and Gender.

In [59]:
```
### [ Step.1 ] Data loading

dateData<-read.delim("LooksOrPersonality.dat", header = TRUE)
speedData<-reshape2::melt(dateData, id = c("participant","gender"),
            measured = c("att_high", "av_high", "ug_high",
                         "att_some", "av_some", "ug_some",
                         "att_none", "av_none", "ug_none"))
names(speedData)<-c("participant", "gender", "groups", "dateRating")

speedData$personality<-gl(3, 60, labels = c("Charismatic", "Average", "Dullard'
speedData$looks<-gl(3,20, 180, labels = c("Attractive", "Average", "Ugly"))
speedData<-speedData[order(speedData$participant),]

speedData %>% head(5)
```

A data.frame: 5 × 6

	participant	gender	groups	dateRating	personality	looks
	<fct>	<fct>	<fct>	<int>	<fct>	<fct>
1	P01	Male	att_high	86	Charismatic	Attractive
21	P01	Male	av_high	84	Charismatic	Average
41	P01	Male	ug_high	67	Charismatic	Ugly
61	P01	Male	att_some	88	Average	Attractive
81	P01	Male	av_some	69	Average	Average

In [60]:
```
### [ Step.2 ] Check assumptions

options(digits = 3)

#----- (a) Normality -----#
speedData %>%
    group_by(gender, personality, looks) %>%
    shapiro_test(dateRating)

#----- (b) Outliers -----#
speedData %>%
    group_by(gender, personality, looks) %>%
    identify_outliers(dateRating)

#----- (c) Homogeneity of Variance -----#
speedData %>%
    group_by(personality, looks) %>%
    levene_test(dateRating ~ gender)
```

A tibble: 18 × 6

gender	personality	looks	variable	statistic	p
<fct>	<fct>	<fct>	<chr>	<dbl>	<dbl>
Female	Charismatic	Attractive	dateRating	0.939	0.5474
Female	Charismatic	Average	dateRating	0.863	0.0827
Female	Charismatic	Ugly	dateRating	0.974	0.9276
Female	Average	Attractive	dateRating	0.907	0.2637
Female	Average	Average	dateRating	0.970	0.8933
Female	Average	Ugly	dateRating	0.959	0.7724
Female	Dullard	Attractive	dateRating	0.952	0.6975
Female	Dullard	Average	dateRating	0.921	0.3642
Female	Dullard	Ugly	dateRating	0.936	0.5104
Male	Charismatic	Attractive	dateRating	0.871	0.1026
Male	Charismatic	Average	dateRating	0.930	0.4482
Male	Charismatic	Ugly	dateRating	0.989	0.9947
Male	Average	Attractive	dateRating	0.946	0.6230
Male	Average	Average	dateRating	0.950	0.6655
Male	Average	Ugly	dateRating	0.965	0.8454
Male	Dullard	Attractive	dateRating	0.989	0.9953
Male	Dullard	Average	dateRating	0.871	0.1013
Male	Dullard	Ugly	dateRating	0.934	0.4890

A tibble: 19 × 8

gender	personality	looks	participant	groups	dateRating	is.outlier	is.extreme
<fct>	<fct>	<fct>	<fct>	<fct>	<int>	<lgl>	<lgl>
Female	Charismatic	Average	P13	av_high	100	TRUE	FALSE
Female	Charismatic	Average	P18	av_high	69	TRUE	TRUE
Female	Dullard	Attractive	P18	att_none	45	TRUE	FALSE
Female	Dullard	Average	P19	av_none	53	TRUE	FALSE
Female	Dullard	Average	P20	av_none	39	TRUE	FALSE
Female	Dullard	Ugly	P11	ug_none	52	TRUE	FALSE
Female	Dullard	Ugly	P16	ug_none	40	TRUE	FALSE
Male	Charismatic	Attractive	P05	att_high	80	TRUE	FALSE
Male	Charismatic	Attractive	P06	att_high	80	TRUE	FALSE
Male	Charismatic	Attractive	P08	att_high	100	TRUE	TRUE
Male	Charismatic	Average	P04	av_high	69	TRUE	FALSE
Male	Charismatic	Average	P08	av_high	94	TRUE	FALSE
Male	Charismatic	Average	P09	av_high	74	TRUE	FALSE
Male	Average	Attractive	P03	att_some	99	TRUE	FALSE
Male	Average	Average	P06	av_some	63	TRUE	FALSE
Male	Average	Ugly	P09	ug_some	58	TRUE	FALSE
Male	Dullard	Average	P09	av_none	38	TRUE	TRUE
Male	Dullard	Ugly	P04	ug_none	53	TRUE	FALSE
Male	Dullard	Ugly	P10	ug_none	39	TRUE	FALSE

A tibble: 9 × 6

personality	looks	df1	df2	statistic	p
<fct>	<fct>	<int>	<int>	<dbl>	<dbl>
Charismatic	Attractive	1	18	1.14846	0.298
Charismatic	Average	1	18	0.05043	0.825
Charismatic	Ugly	1	18	0.00476	0.946
Average	Attractive	1	18	0.45236	0.510
Average	Average	1	18	1.81610	0.194
Average	Ugly	1	18	0.11917	0.734
Dullard	Attractive	1	18	1.67482	0.212
Dullard	Average	1	18	0.02133	0.886
Dullard	Ugly	1	18	0.07996	0.781

In [61]:
```
### [ Step.3-1 ] ANOVA

(speedModel<-anova_test(data = speedData, dv = dateRating,
            wid = participant,  between = gender,
            within = c(looks, personality), detailed = TRUE))

#speedModel.aov<-aov(dateRating ~ (gender * looks * personality) +
#    Error(participant/(looks * personality)), data = speedData)
#summary(speedModel.aov)
```

$ANOVA

A data.frame: 8 × 9

Effect	DFn	DFd	SSn	SSd	F	p	p<.05	ges
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
(Intercept)	1	18	846249.8	760	2.00e+04	7.01e-29	*	9.94e-01
gender	1	18	0.2	760	5.00e-03	9.46e-01		4.07e-05
looks	2	36	20779.6	883	4.24e+02	9.59e-26	*	8.09e-01
personality	2	36	23233.6	1274	3.28e+02	7.69e-24	*	8.26e-01
gender:looks	2	36	3944.1	883	8.04e+01	5.23e-14	*	4.45e-01
gender:personality	2	36	4420.1	1274	6.24e+01	1.97e-12	*	4.74e-01
looks:personality	4	72	4055.3	1993	3.66e+01	1.10e-16	*	4.52e-01
gender:looks:personality	4	72	2669.7	1993	2.41e+01	1.11e-12	*	3.52e-01

$`Mauchly's Test for Sphericity`

A data.frame: 6 × 4

	Effect	W	p	p<.05
	<chr>	<dbl>	<dbl>	<chr>
1	looks	0.960	0.708	
2	gender:looks	0.960	0.708	
3	personality	0.929	0.536	
4	gender:personality	0.929	0.536	
5	looks:personality	0.613	0.534	
6	gender:looks:personality	0.613	0.534	

$`Sphericity Corrections`

A data.frame: 6 × 9

Effect	GGe	DF[GG]	p[GG]	p[GG]<.05	HFe	DF[HF]	p[HF]	p[HF]<.05
<chr>	<dbl>	<chr>	<dbl>	<chr>	<dbl>	<chr>	<dbl>	<chr>
looks	0.962	1.92, 34.62	7.62e-25	*	1.074	2.15, 38.68	9.59e-26	*
gender:looks	0.962	1.92, 34.62	1.49e-13	*	1.074	2.15, 38.68	5.23e-14	*
personality	0.934	1.87, 33.62	2.06e-22	*	1.038	2.08, 37.37	7.69e-24	*
gender:personality	0.934	1.87, 33.62	9.44e-12	*	1.038	2.08, 37.37	1.97e-12	*
looks:personality	0.799	3.2, 57.55	9.00e-14	*	0.992	3.97, 71.44	1.43e-16	*
gender:looks:personality	0.799	3.2, 57.55	1.47e-10	*	0.992	3.97, 71.44	1.34e-12	*

In [62]:
```
### [ Step.3-2 ] Post-hoc test

#-> 1. Simple 2-way interaction: Estimation by different gender
two.way <- speedData %>%
    group_by(gender) %>%
    anova_test(dv = dateRating, wid = participant, within = c(looks, personalit
get_anova_table(two.way) %>%
    adjust_pvalue(method='bonferroni')
```

A tibble: 6 × 9

gender	Effect	DFn	DFd	F	p	p<.05	ges	p.adj
<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>
Female	looks	2.00	18.0	66.4	4.90e-09	*	0.559	2.94e-08
Female	personality	2.00	18.0	274.9	3.23e-14	*	0.902	1.94e-13
Female	looks:personality	2.32	20.8	34.9	1.02e-07	*	0.563	6.12e-07
Male	looks	2.00	18.0	443.6	4.86e-16	*	0.903	2.92e-15
Male	personality	2.00	18.0	68.3	3.94e-09	*	0.618	2.36e-08
Male	looks:personality	4.00	36.0	26.9	2.24e-10	*	0.593	1.34e-09

In [63]:
```
### [ Step.3-2 ] Post-hoc test

#-> 2. Simple simple main effect:
#--> (a) estimate Looks effect at each personality
look.effect <- speedData %>%
    group_by(gender, personality) %>%
    anova_test(dv = dateRating, wid = participant, within = looks)
get_anova_table(look.effect) %>%
    adjust_pvalue(method='bonferroni')

#--> (b) estimate Personality effect at each Looks
personality.effect <- speedData %>%
    group_by(gender, looks) %>%
    anova_test(dv = dateRating, wid = participant, within = personality)
get_anova_table(personality.effect) %>%
    adjust_pvalue(method='bonferroni')
```

A tibble: 6 × 10

gender	personality	Effect	DFn	DFd	F	p	p<.05	ges	p.adj
<fct>	<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>
Female	Charismatic	looks	2	18	0.64	5.39e-01		0.032	1.00e+00
Female	Average	looks	2	18	105.03	1.19e-10	*	0.865	7.14e-10
Female	Dullard	looks	2	18	10.17	1.00e-03	*	0.371	6.00e-03
Male	Charismatic	looks	2	18	82.58	8.55e-10	*	0.846	5.13e-09
Male	Average	looks	2	18	121.27	3.59e-11	*	0.918	2.15e-10
Male	Dullard	looks	2	18	292.57	1.88e-14	*	0.953	1.13e-13

A tibble: 6 × 10

gender	looks	Effect	DFn	DFd	F	p	p<.05	ges	p.adj
<fct>	<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>
Female	Attractive	personality	2	18	124.887	2.80e-11	*	0.907	1.68e-10
Female	Average	personality	2	18	115.487	5.40e-11	*	0.889	3.24e-10
Female	Ugly	personality	2	18	260.008	5.25e-14	*	0.941	3.15e-13
Male	Attractive	personality	2	18	0.111	8.96e-01		0.010	1.00e+00
Male	Average	personality	2	18	128.133	2.26e-11	*	0.892	1.36e-10
Male	Ugly	personality	2	18	12.124	4.63e-04	*	0.498	2.78e-03

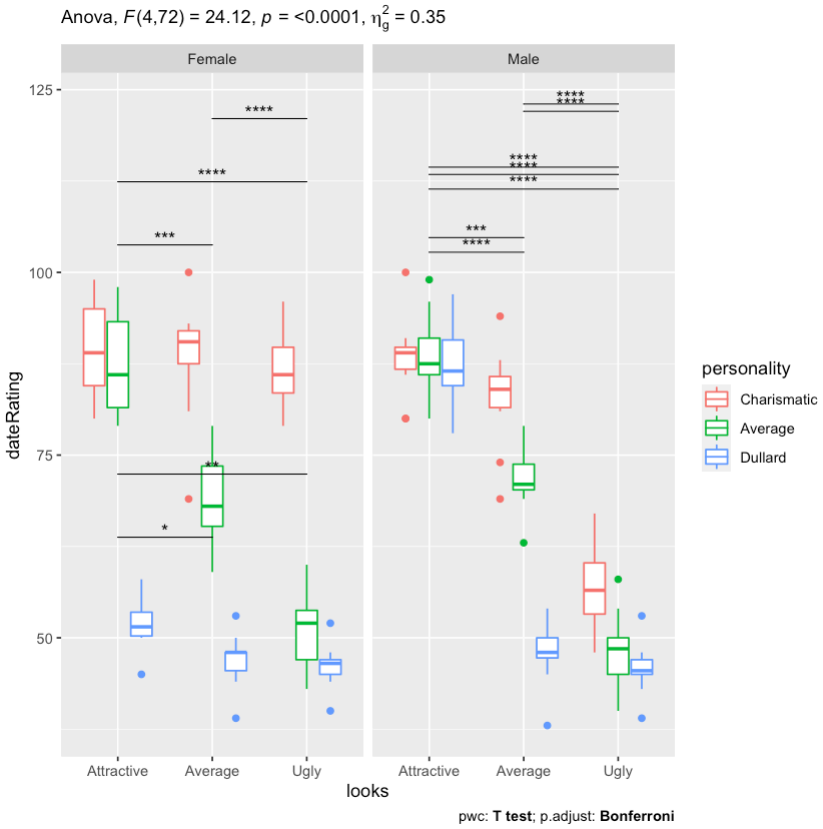
In [64]:
```
### [ Step.3-2 ] Post-hoc test

#-> 3. Simple simple pairwise comparisons: drink effects at each emotion
(pwc <- speedData %>%
    group_by(gender, personality) %>%
    pairwise_t_test(dateRating ~ looks, paired = TRUE, p.adjust.method="bonferr
```

A rstatix_test: 18 × 12

	gender	personality	.y.	group1	group2	n1	n2	statistic	df	p	p.adj	p.ad
	<fct>	<fct>	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	
1	Female	Charismatic	dateRating	Attractive	Average	10	10	0.372	9	7.19e-01	1.00e+00	
2	Female	Charismatic	dateRating	Attractive	Ugly	10	10	1.710	9	1.21e-01	3.63e-01	
3	Female	Charismatic	dateRating	Average	Ugly	10	10	0.661	9	5.25e-01	1.00e+00	
4	Female	Average	dateRating	Attractive	Average	10	10	5.835	9	2.48e-04	7.44e-04	
5	Female	Average	dateRating	Attractive	Ugly	10	10	18.666	9	1.67e-08	5.01e-08	
6	Female	Average	dateRating	Average	Ugly	10	10	7.932	9	2.37e-05	7.11e-05	
7	Female	Dullard	dateRating	Attractive	Average	10	10	3.560	9	6.00e-03	1.80e-02	
8	Female	Dullard	dateRating	Attractive	Ugly	10	10	4.934	9	8.09e-04	2.00e-03	
9	Female	Dullard	dateRating	Average	Ugly	10	10	0.582	9	5.75e-01	1.00e+00	
10	Male	Charismatic	dateRating	Attractive	Average	10	10	2.332	9	4.50e-02	1.34e-01	
11	Male	Charismatic	dateRating	Attractive	Ugly	10	10	12.340	9	6.07e-07	1.82e-06	
12	Male	Charismatic	dateRating	Average	Ugly	10	10	8.924	9	9.15e-06	2.75e-05	
13	Male	Average	dateRating	Attractive	Average	10	10	5.855	9	2.42e-04	7.26e-04	
14	Male	Average	dateRating	Attractive	Ugly	10	10	15.896	9	6.81e-08	2.04e-07	
15	Male	Average	dateRating	Average	Ugly	10	10	9.887	9	3.93e-06	1.18e-05	
16	Male	Dullard	dateRating	Attractive	Average	10	10	20.802	9	6.42e-09	1.93e-08	
17	Male	Dullard	dateRating	Attractive	Ugly	10	10	19.432	9	1.17e-08	3.51e-08	
18	Male	Dullard	dateRating	Average	Ugly	10	10	1.142	9	2.83e-01	8.49e-01	

In [21]:
```
### [ Step.4 ] Effect size -- Generalized Eta Squared
```

In [66]:
```
### [ Step.5 ] Visualization

pwc <- pwc %>% add_xy_position(x="looks")
#pwc.filter <- pwc %>% filter(gender != 'Male' & group1 != 'Attractive')
ggplot(speedData, aes(looks, dateRating)) +
    geom_boxplot(aes(color=personality))+ facet_grid(. ~ gender) +
    stat_pvalue_manual(pwc, tip.length = 0, hide.ns=TRUE) +
    labs(subtitle = get_test_label(speedModel, detailed = TRUE),
         caption = get_pwc_label(pwc))
```



~ *Report* ~

- **[Interaction]** A 3-way Mixed ANOVA was performed. The attractiveness of the date were significant interacted with the gender, the personality level and the looks of the date ($\boldsymbol{F_{4,\,72} = 24.1,\ p < .001,\ \eta^2 = .35}$).
- **[Post-hoc]** There were significant simple two-way interactions (Looks × Charisma) for both male ($\boldsymbol{F_{4,\,36} = 26.9,\ p < .001}$) and female ($\boldsymbol{F_{2.3,\,20.8} = 34.9,\ p < .001}$). However, the simple simple main effect showed different trends of preferences between gender. If the date was with charismatic personality, women showed similar preference on the attractiveness of the date ($F_{2,\,18} = 0.64,\ p = 1$). Instead, men prioritize physical appearance over charisma ($F_{2,\,18} = 0.11,\ p = 1$).